

Lab 1 — Setting up Google Colab

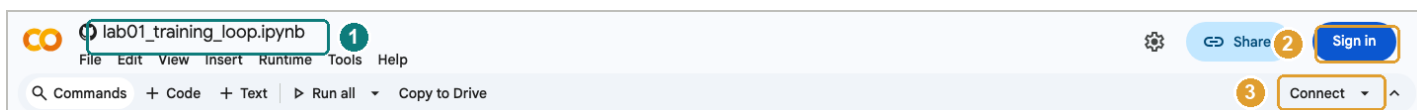
Everything runs in your browser. Nothing to install.

What you need: a web browser and a **free Google account** (a personal @gmail.com is fine — some hospital and university accounts block Colab, so if yours does, use a personal one). **You do not need:** Python, PyTorch, an install, or a GPU. *Colab* is a free Google service that runs Python notebooks on Google's computers; your laptop only shows the page.

Step 1 — Open the Lab 1 notebook

Click the Lab 1 link (on the course page, or type it in). It opens Colab with the notebook already loaded — there is nothing to download first.

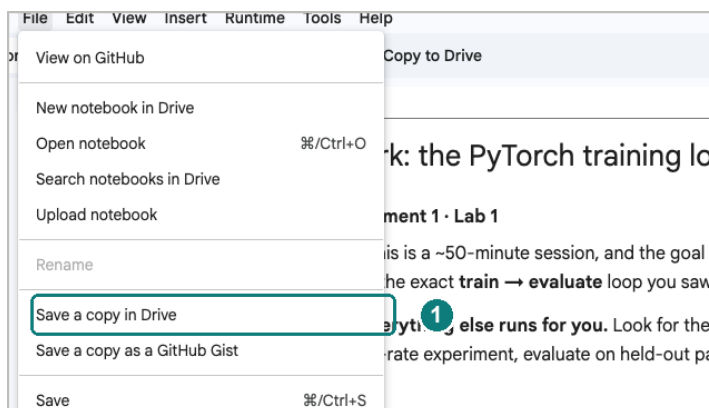
https://colab.research.google.com/github/indigobio/msacl_dl_course/blob/main/student_pack/labs/lab01_training_loop.ipynb



- (1) Check the file name reads `lab01_training_loop.ipynb` — that is the right notebook.
- (2) **Sign in** (top right) with your Google account. Until you do, you can read the notebook but not run it.
- (3) **Connect** (top right) starts your free machine. It usually connects on its own the first time you run a cell; if it says *Connect*, click it and wait for a green tick.

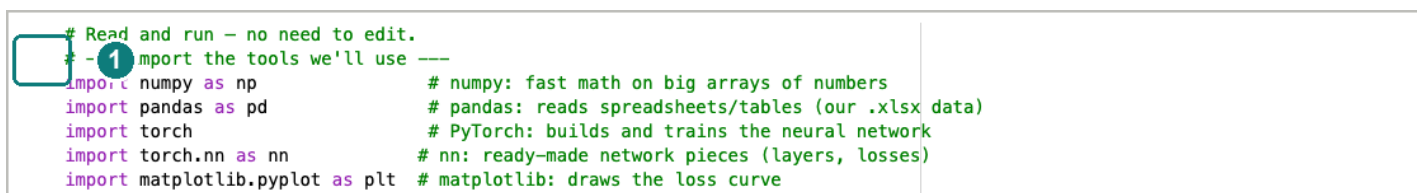
Step 2 — Save your own copy (do this before you type anything)

The link opens a *read-only* copy. Use **File** → **Save a copy in Drive** and work in that copy, or your edits will be lost when you close the tab. Colab opens the copy in a new tab, named *Copy of lab01_training_loop.ipynb* — **that is the tab you want to work in**.



Step 3 — Run a cell

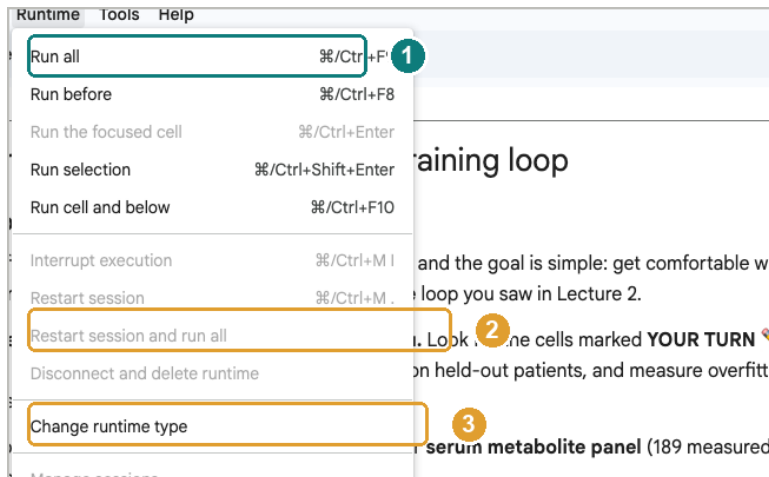
A notebook is a list of *cells*. A cell holds either text or code. Click a code cell, then press **Shift + Enter** — or hover over the (1) bracket at its left, which turns into a ► button, and click it.



- Run the cells **in order, top to bottom**. Later cells depend on earlier ones.
- While a cell runs the bracket shows a spinner; when it finishes you get a number, e.g. [1].
- **Run the first code cell, “Setup”, before anything else**. It installs the course packages and downloads the lab's data for you — about a minute, ending with `lab01 is ready`. You never download or copy data files yourself.
- Only the cells marked **YOUR TURN** (the pencil icon) need editing. Run all the others exactly as they are.

Step 4 — If something gets tangled

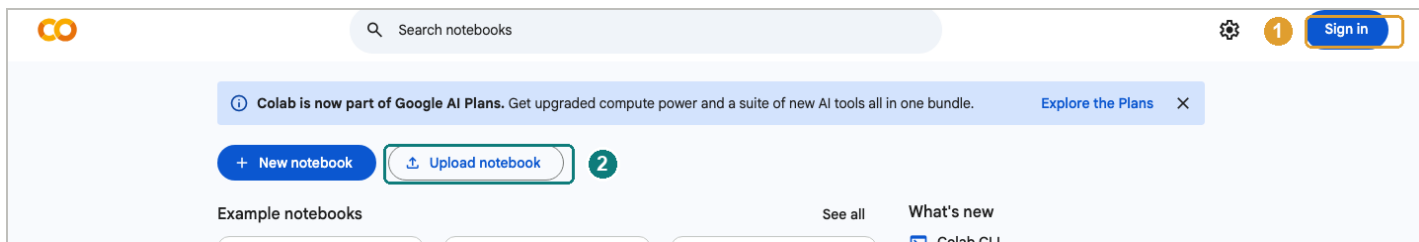
The **Runtime** menu is the fix for almost everything.



- (1) **Run all** — runs the whole notebook from the top. Good after you fix a cell.
- (2) **Restart session and run all** — throws away every variable and starts clean. This is the answer to “it worked a minute ago”. You lose nothing but the running state; your typed code stays.
- (3) **Change runtime type** — where you would pick a **T4 GPU**. **Lab 1 does not need one**; the default CPU is plenty and connects faster. You will use this in Lab 2.

Step 5 — If the link does not open

Go to colab.research.google.com directly, sign in (1), then use **File** → **Open notebook** → **GitHub** and paste [indigobio/msacl_dl_course](https://github.com/indigobio/msacl_dl_course). Or use **Upload notebook** (2) if an instructor hands you the .ipynb file on a USB stick.



When things go wrong

What you see

What to do

“Sign in” keeps reappearing, or Colab will not load

Your institutional Google account may block Colab. Sign in with a personal @gmail.com instead.

The setup cell says “Restarting now”, then “session restarted” / “crashed”

Expected, once per session: the course packages replaced ones Colab had already loaded. Wait for it to reconnect, then run the **setup cell** again.

`AttributeError` mentioning `numpy` or `_multiarray_umath`
`NameError / ... is not defined`

The session was not restarted after setup. **Runtime** → **Restart session**, then run the setup cell again.

You skipped a cell, or ran them out of order. **Runtime** → **Restart session and run all**.

A cell has been running for minutes

Normal for training cells. If it is stuck, **Runtime** → **Interrupt execution**, then restart and run all.

“Cannot connect to runtime” / disconnected

Free Colab disconnects after a while idle. Click **Connect** and **Runtime** → **Run all**; you lose only the running state.

An **assert** cell failed

That is the notebook checking your answer — it is meant to catch you. Read the message, fix the marked line, re-run that cell. The **Code Hint Sheet** lists the options for every blank.

You edited the wrong tab and lost your work

You were in the read-only original. Redo **File** → **Save a copy in Drive** and check the tab title starts with Copy of.

Screenshots are of the Google Colab interface, captured 22 September 2026, with callout numbers added for this sheet. Colab is a Google product and its appearance changes from time to time — if a button has moved, the menu path in the text is still the one to follow. A clickable version of this sheet is [lab01_colab_setup.html](#).